MILITARY SPECIFICATION

INVERTER, POWER, STATIC: 10516412

This amendment forms a part of Military Specification MIL-I-60166(MU), dated 14 May 1964.

Paragraph 2.1: Delete:

"MIL-I-26600 Interference Control Requirements, Aeronautical Equipment"

and substitute

MIL-E-55301 Electromagnetic Compatability".

Paragraph 3.4: Add "(see 6.2(d))".

Paragraph 3.5.2: Delete last sentence and substitute:

"The inverter shall meet the requirements of 3.7 before and after vibration."

Paragraph 3.5.2.1: Add the following new paragraph:

"Monitoring during vibration.- The inverter input voltage shall be 24 vdc, a load of 150 volt amperes, and a unity power factor. The characteristics to be monitored are as follows: input current, voltage output, output current, frequency of output voltage, and the harmonic content of the output waveform. All monitored values shall meet the requirements of 3.7."

Paragraph 3.7: Delete first sentence and substitute the following:

"Before and after vibration (see 3.5.2) the inverter shall meet the requirements of 3.7.1 thru 3.7.6 without spasmodic or intermittent operation, with input voltages of 18 vdc, 24 vdc and 30 vdc."

Paragraph 3.7.5: Delete and substitute the following:

"3.7.5 Harmonic distortion.- The inverter output shall be free of all electrical voltage spikes at the output during switching and operating at all combinations of output loads. The inverter shall have a harmonic

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distortion of less than 4% for all loads and all power factor ranges between 0.8 leading and 0.6 lagging. With a unity power factor and at a load of 75 VA, the maximum variation of harmonic distortion shall be 1.5%."

Paragraph 3.10: Delete "MIL-I-26600 (Class III)" and substitute "MIL-E-55301 at a frequency range of 1.5 megacycles and at a level of 83 db/luv/mc."

Paragraph 4.3.2, Table II: Delete the following:

"115. Radio interference 3.10 4.6.11"

Paragraph 4.4.1: Add the following control test:

"Radio interference 3.10 4.6.11"

Paragraph 4.4.3, Reliability: Delete in its entirety.

Paragraph 4.6.5: Add the following sentence:

"This test shall only be conducted for Qualification (QPL) (see 4.2.1.2) and as otherwise specified in the contract (see 6.2(d))."

Paragraph 4.6.6.2: Delete the second sentence and substitute the following:

"Before and after vibration, the inverter shall be tested at inputs 18 vdc, 24 vdc and 30 vdc in accordance with 4.6.8 to 4.6.8.6, inclusive to determine compliance with 3.7. During vibration in each of the three mutually perpendicular planes (see 3.5.2), the inverter shall be monitored when energized at 150 VA, 24 vdc and unity power factor to determine compliance with 3.5.2.1."

Paragraph 4.6.8.6, third sentence: Delete "for a minimum of one hour" and substitute "and be stabilized for a period of five minutes."

Paragraph 4.6.11: Delete "MIL-I-26600" and substitute "MIL-E-55301". Add the following sentence "The test for requirement chassis, case and mounting continuity specified in MIL-E-55301 shall be conducted as an in-process test prior to application of paint finish on the exterior surface."

Paragraph 6.2: Add the following sub-paragraph:

"(d) Compliance with the Reliability requirement and test (see 3.4 and 4.6.5)."

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